Basic Network Sniffer

CodeAlpha Cyber Security Internship

Submitted By

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Domain	Cyber Security
Programming Language	Python

Objective

The objective of this project was to develop a basic network sniffer capable of capturing live network packets and displaying useful packet information such as source IP address, destination IP address, protocol type, source port, destination port, packet number, and timestamp.

Introduction

A network packet sniffer is a tool used to capture and analyze packets travelling across a network. It helps network administrators and cybersecurity professionals monitor network communication, troubleshoot connectivity issues, and understand how devices communicate.

For this project, Python and the Scapy library were used to build a simple packet sniffer capable of capturing live network traffic.

Tools Used

- Python
- Scapy
- Npcap
- Visual Studio Code

Implementation

The application captures five live packets from the network and displays:

- Packet Number
- Timestamp
- Source IP Address
- Destination IP Address
- Protocol
- Source Port
- Destination Port

The Scapy library was used to sniff packets, while Python's datetime module was used to display the capture time.

Features

- Live Packet Capture
- Packet Numbering
- Timestamp
- IP Address Detection

- Protocol Detection
- Port Number Detection
- Simple Console Interface

Learning Outcomes

During this project, I learned:

- How network packets travel
- Difference between Source IP and Destination IP
- TCP, UDP and ICMP protocols
- Importance of Port Numbers
- Packet Sniffing using Scapy
- Python Networking Basics
- Network Monitoring Fundamentals

Conclusion

This project successfully demonstrated the fundamentals of packet sniffing using Python and Scapy. It provided practical experience in capturing and analyzing live network traffic while improving my understanding of networking concepts and cybersecurity fundamentals.